COMP 551 – MiniProject 1 Report (Group 15)

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Abstract

This project studies daily bike rental demand using the UCI Bike Sharing Dataset (`day.csv`) and linear regression. We first explored the dataset, checking for missing values, removing identifier and leakage variables, and handling categorical features via one-hot encoding. We then performed exploratory data analysis using visualizations to better understand relationships between the remaining variables, which informed our feature engineering choices. We also standardized the continuous weather features in a duplicated dataset. Using a fixed 75/25 train/test split, we implement ordinary least squares regression from scratch with a numerically stable solver and evaluate performance using mean squared error (MSE) on both training and test sets. We then examine the effect of simple non-linear feature engineering on model fit, focusing on changes in training error to illustrate how feature transformations increase model capacity and the limitations of linear regression for modeling demand patterns in this dataset.

1 Introduction

The UCI Bike Sharing Dataset contains daily aggregated records of bike rentals, calendar information (e.g., season, working day, date), and weather information such as temperature, humidity, and windspeed. In this project, we use the `day.csv` file and model the target variable `cnt`, which represents the total number of rented bikes per day. Weather features are continuous, while calendar features are discrete or categorical, which motivates careful preprocessing before fitting a regression model. Our modeling approach is ordinary least squares (OLS) linear regression, implemented from scratch using the closed-form solution and evaluated using mean squared error on both the training and test sets. The goals of this assignment are (i) to practice data cleaning and feature representation choices (preventing data leakage, one-hot encoding categorical variables, and optionally standardizing continuous variables), (ii) to establish a baseline linear model and evaluate its performance, and (iii) to explore whether simple non-linear feature engineering can reduce training error and increase the model's ability to fit the data.

2 Data Preprocessing and Exploration

We begin by loading the daily aggregated Bike Sharing dataset (`day.csv`) and inspecting its structure, feature types, and size. A first pass through the data confirms that there are no missing or malformed values, so no imputation or row removal is necessary. We checked both for `NaN`, `None`, and empty strings/variables using the code shown below.

```
print("missing values (NaN or None):")
print(raw_data.isna().sum().sum())

print("empty values")
print((raw_data == "").sum())
```

Listing 1: Missing variable check

We then remove several features that are either identifiers or would introduce data leakage, namely `instant`, `dteday`, `casual`, and `registered`, since the latter two directly sum to the target variable `cnt`.

Categorical calendar and weather-related variables (`season`, `mnth`, `weekday`, and `weathersit`) are transformed using one-hot encoding, with the first category dropped to avoid multicollinearity. All resulting boolean features are converted to integer values to ensure numerical compatibility with linear regression.

To better understand the data, we visualized the distribution of the target variable and continuous features using histograms, and examined relationships between features and the target through scatter

plots and box plots across calendar and weather-related categories, as well as a correlation heatmap for continuous features. These visualizations highlight differences in demand across seasons (Figure 1e) and weather conditions (Figure 1c), as well as approximate linear trends between temperature-related features (`temp` and `atemp`) and daily bike rentals (Figure 1a and 1b).

Exploratory data analysis also reveals certain weather variables are correlated (Figure 1d), particularly temperature and perceived temperature, which motivates careful feature selection and later feature engineering. While correlation alone does not imply importance, these observations provide useful intuition for constructing and interpreting the regression model.

3 Methods

We use ordinary least squares linear regression as our modeling approach and implement it from scratch using NumPy. A bias term is added to the design matrix, and the model parameters are estimated using a numerically stable least-squares solver (`numpy.linalg.lstsq`), which avoids explicit matrix inversion and improves numerical stability. Model performance is evaluated using mean squared error (MSE) on both training and test sets:

$$\mathrm{MSE}(\hat{y}, y) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2.$$

The dataset is randomly shuffled with a fixed seed for reproducibility and split into a 75% training set and a 25% test set. To study the effect of feature scaling, we create both a non-standardized and a standardized version of the dataset. Standardization is applied only to continuous weather-related features using statistics computed on the training set and then applied to the test set to avoid data leakage.

This separation between training and test preprocessing ensures that model evaluation reflects generalization performance rather than artifacts of information leakage.

In addition to the baseline model, we perform simple non-linear feature engineering on selected continuous variables to increase the model capacity. Polynomial terms are added for perceived temperature (`atemp`) and humidity to allow the model to capture curved relationships between weather conditions and bike demand. We include squared terms for perceived temperature (`atemp`) and humidity, and a cubic term for the perceived temperature to better capture uneven behavior at extreme weather values, since the drop in bike rentals during very hot days does not mirror the drop during very cold days. We also added interaction terms between temperature, humidity, and windspeed to model cases where the effect of one variable depends on another, which cannot be represented by our basis linear model. Lastly, we applied a logarithmic transformation of windspeed to account for diminishing effects at higher wind speeds.

4 Results

Table 1 summarizes our MSE results obtained for the baseline linear regression model and the feature-engineered model on both the training and test sets. For the baseline model, performance is similar with and without continuous feature standardization, but improves numerical stability and produces more interpretable coefficient magnitudes.

Table 1: Mean Squared Error (MSE) for baseline and feature-engineered linear regression models.

Model	Train MSE	Test MSE
Baseline Linear Regression	407,345	516,802
Feature-Engineered Linear Regression	406,244	516,789

We can see that by adding non-linear feature transformations leads to a small reduction in training error, with the MSE for the training datasets decreasing from 407,345 to 406,244, approximatly a 0.27% reduction. This decrease indicates that the engineered features allow the model to better fit the training data by capturing non-linear relationships between bike demand and the weather variables. It's also good

to note that these values are identical for both the standardized and non-standardized versions of the dataset. The test MSE does not necessarily improve to the same extent, which is consistent with the increased risk of overfitting introduced by additional features.

These results confirm our expectations for linear regression. Although feature engineering increases the model's ability to fit the training data, the improvements don't necessarily translate to better generalization.

5 Discussion and Conclusion

This project illustrates the importance of preprocessing and feature representation when applying linear regression to real-world datasets. A valid learning setup requires removing identifier and leakage-prone variables, and one-hot encoding enables the incorporation of categorical variables into a linear model. Exploratory data analysis provides useful intuition regarding feature relevance and correlations with the target variable.

Our findings show that standardizing continuous features improves numerical stability without meaningfully changing predictive performance. Introducing non-linear feature transformations reduces training error, demonstrating how linear regression can model more complex patterns when paired with suitable feature engineering. However, the fact that test performance does not always improve highlights the limitations of linear regression and the increased risk of overfitting with additional features.

Something we thought might be a limitation of this project is the use of a random split of the data to create our train-test sets. This ignores the temporal structure of the data and since bike demand shows strong seasonal and time-dependent patterns, the random shuffling may lead to optimistic estimates of performance. A natural next step would to use a time based split, training the model on earlier dates and evaluating on later periods, to better reflect forecasting scenarios and more accurately calculate perfomance on future bike demand.

Overall, the results showed that improved training performance alone is not sufficient evidence of better generalization, and evaluation on held-out data remains essential.

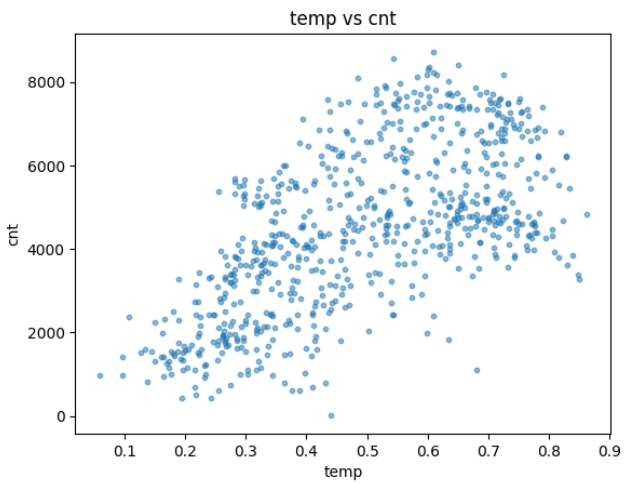
Statement of Contributions

The team met and decided on the task split as follows: Louis Stalet handled Tasks 1 and 2, Mark Maroun handled Task 3, and Rami El-Nawam wrote the report (Task 4). All members reviewed the final submission and discussed results and conclusions.

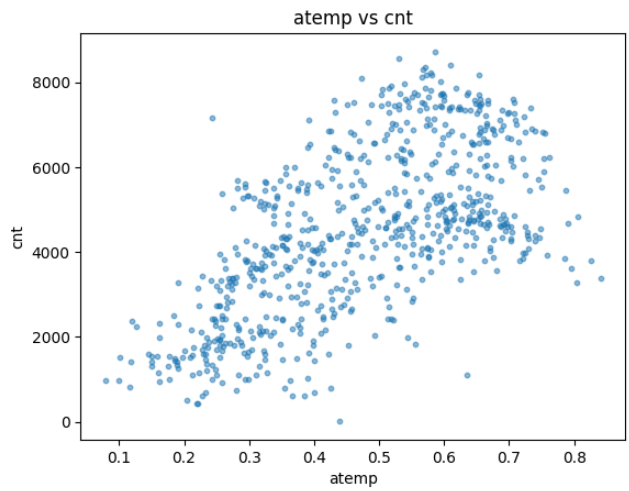
Statement on the Use of LLMs and Other Resources

We used ChatGPT to help reformat and polish the report into LaTeX and to improve clarity. We also used it to generate our README.md file. We did not use it to generate code or derive results.

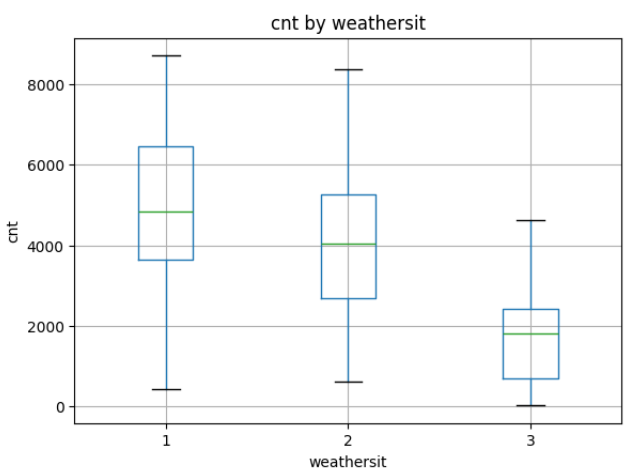
Exploratory Figures



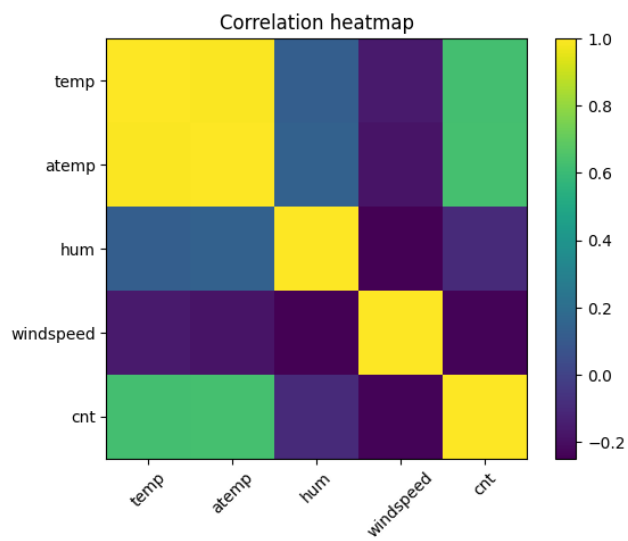
(a) Temperature vs. rentals



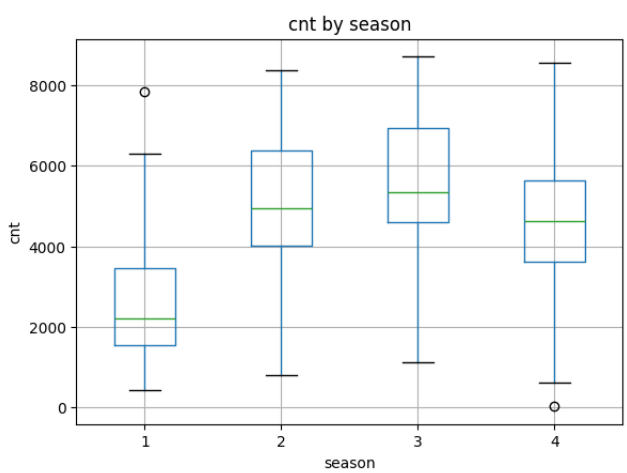
(b) Perceived temperature vs. rentals



(c) Bike rentals per weather type



(d) Correlation Heatmap



(e) Bike rentals per seasons

Figure 1: Exploratory analysis of key weather variables referenced throughout the report.